

Fundamental concept of boolean algebra

Only two mathematical operations

Addition

OR

Conjunction (logical addition)

The '+' symbol is read as x or y

$$\begin{aligned} x + y \\ 0 + 0 &= 0 \\ 0 + 1 &= 1 \\ 1 + 0 &= 1 \\ 1 + 1 &= 1 \end{aligned}$$

Multiplication (Repeated addition)

AND

Disjunction (Logical multiplication)

The dot symbol is read as x and y

$$\begin{aligned} x \cdot y \\ 0 \cdot 0 &= 0 \\ 0 \cdot 1 &= 0 \\ 1 \cdot 0 &= 0 \\ 1 \cdot 1 &= 1 \end{aligned}$$

1 and 0,
high and low,
true and false,
on and off.

George Boole, has written a book "An Investigation of the laws of thought". This book is based on logic and Probabilities.

Later on Claude E. Shannon at Massachusetts Institute of Technology wrote a thesis. Today Boolean Algebra is the back bone of Computer Circuit analysis.

Three basic laws of Boolean Algebra are same as normal algebra.

1. Commutative law:- $A + B = B + A$, $A \cdot B = B \cdot A$

2. Associative law :- $(A + B) + C = A + (B + C)$, $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

3. Distributive law :-
i) $A(B + C) = AB + AC$
ii) $A + B \cdot C = (A + B)(A + C)$

x = variables
 y = stores decimal numbers in normal algebra.
But in boolean algebra it will store only binary values

Logic Gates: A logic gate is defined as an electronic circuit

binary values

Logic Gates: A logic gate is defined as an electronic circuit with two or more input signals and have one output signal and it depicts the logic clearly.

There are 3 types of logic Gates :

1. Basic Gates

- OR gate
- AND gate
- Not gate

2. Universal Gates

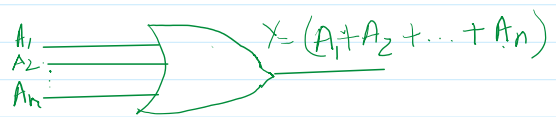
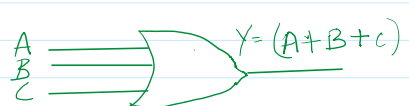
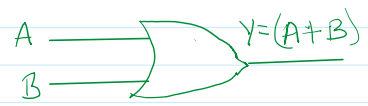
- NOR gate
- NAND gate

3. Exclusive Gates

- XOR gate
- XNOR gate

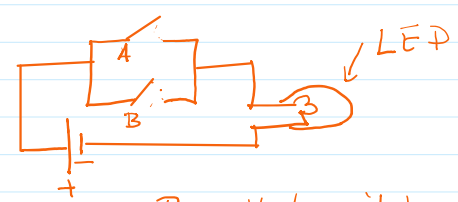
BASIC GATES

1. OR Gate :- An OR Gate is a logic ckt. with '2' or more input signals and '1' output signal. The output signal will be high or 1 or true or on; if any of the input signal is high or 1 or true or on. This is called (logic-1) gate.



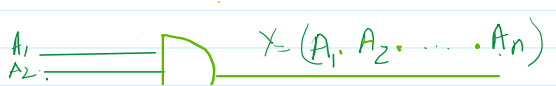
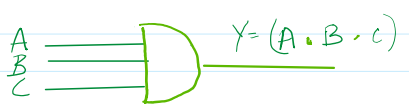
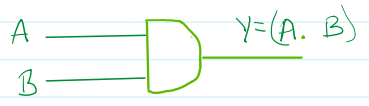
	A	B	Y
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

$2^n = 2^2 = 4$
 ↑
 combination
 (0, 1, 2, 3)



Parallel switch connection

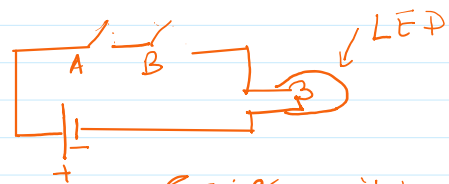
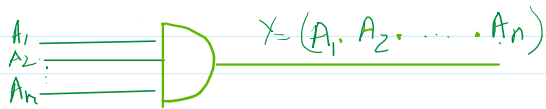
ii) AND Gate :- An AND Gate is a logic ckt. with '2' or more input signals and '1' output signal. The output signal will be high or 1 or true or on; if all of the input signal is high or 1 or true or on. The logic is represented by dot '.' symbol.



	A	B	Y
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

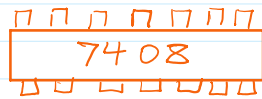
$2^n = 2^2 = 4$
 ↑
 combination
 (0, 1, 2, 3)



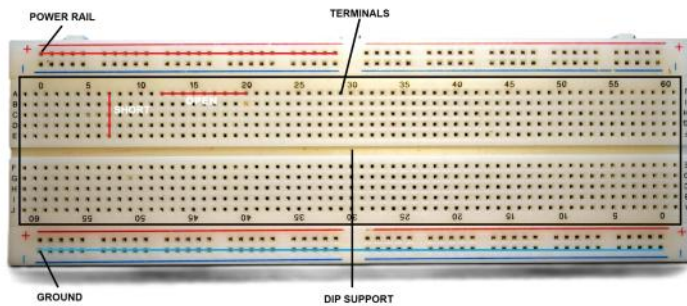
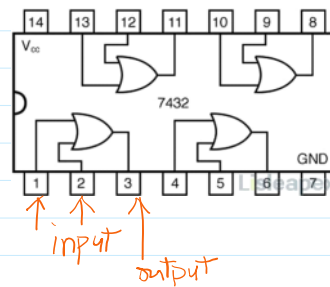
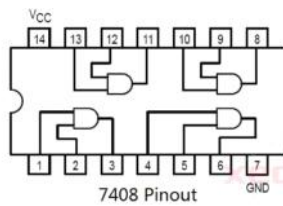
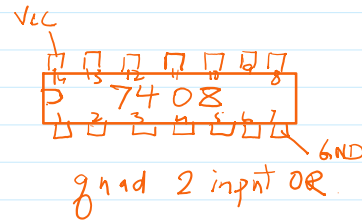


Series switch connection

Integrated ckt. (IC)



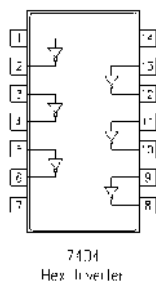
Quad 2 input AND



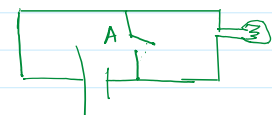
NOT Gate $A \rightarrow \bar{A}$

A inverter or NOT gate is a ckt. that takes one input signal and output signal is always opposite or complement of the input signal

A	$\bar{A}$
0	1
1	0



7404 Hex Inverter



Inverter ckt.
OR
switch ckt.

NOT GATE IC

BASIC DUALITY IN BOOLEAN ALGEBRA

We state the duality theorem in the following:

1. Changing each OR (+) sign to an AND (.) sign.
2. Changing each AND (.) sign to an OR (+) sign.
3. Complement each '0' to '1' or '1' to '0'.

$$A + 0 = A \cdot 1$$

$$A + B \text{ dual is } A \cdot B$$

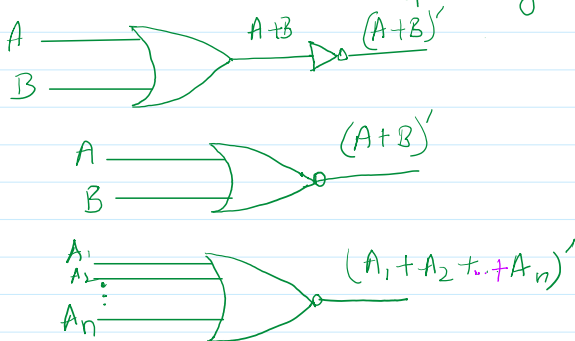
dual relationship $\left\{ \begin{array}{l} A \cdot (B + C) = A \cdot B + A \cdot C \\ (A + B) \cdot C = (A + B) \cdot C \end{array} \right\}$ distributive law

$$1 + A \text{ dual is } 0 \cdot A$$

$$1 + \bar{A} \text{ dual is } 0 \cdot \bar{A}$$

UNIVERSAL Gates (De-Morgan's law)

1. NOR gate: This gate takes two or more input and output is inverter of OR gate.



A	B	(A+B)'
0	0	1
0	1	0
1	0	0
1	1	0

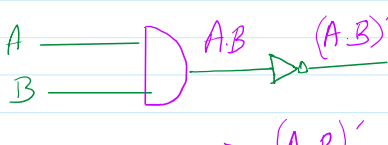
Truth Table of NOR gate

De-Morgan's law

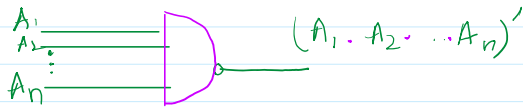
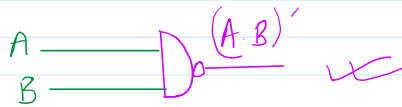
$$(A+B)' = \bar{A} \cdot \bar{B}$$

dual of this expression $(A \cdot B)' = \bar{A} + \bar{B}$

1. NAND gate: This gate takes two or more input and output is inverter of AND gate.



A	B	(A.B)'
0	0	1



A	B	$(A.B)'$
0	0	1
0	1	1
1	0	1
1	1	0

Truth Table of NOR gate

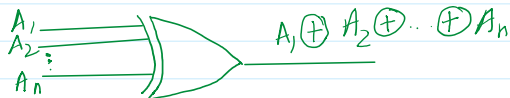
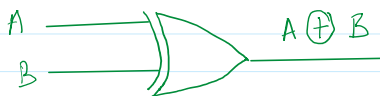
$$(A.B)' = \bar{A} + \bar{B}$$

EXCLUSIVE GATES

1. XOR Gate: It takes 2 or more input signals and output one signal.

The output signal is high or 1 or true, iff we had odd number of high or 1 or true in a row of input signals, otherwise output is low or 0 or false.

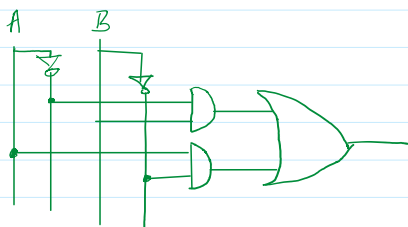
It is represented by $\oplus$ plus in a circle symbol.



A	B	$A \oplus B$	
0	0	0	$A+B$ (Product of Sum) POS
0	1	1	$\bar{A}B$ (Sum of Product) SOP
1	0	1	$A\bar{B}$ (Sum of Product) SOP
1	1	0	$\bar{A} + \bar{B}$ (Product of Sum) POS

POS $\rightarrow A \oplus B = (A+B) \cdot (\bar{A} + \bar{B})$

SOP $\rightarrow A \oplus B = \bar{A}B + A\bar{B}$



XOR gate using basic gates.

SOP Expression.

	A	B	C	$A \oplus B \oplus C$
0	0	0	0	0 ✓
1	0	0	1	1 ✓
2	0	1	0	1 ✓
3	0	1	1	0 ✓
4	1	0	0	1 ✓
5	1	0	1	0 ✓

$$A \oplus B \oplus C = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

POS $\leftarrow$ 1 is barred / complement
0 is selected as output

3	0	1	1	0	✓
4	1	0	0	1	✓
5	1	0	1	0	✓
6	1	1	0	0	
7	1	1	1	1	✓

POS ← 1 is barred / complement
0 is selected as output

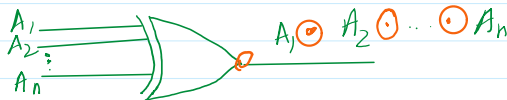
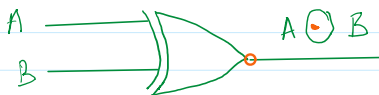
$$A \oplus B \oplus C = (A+B+C)(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)$$

HW. Draw the logical ckt. using basic gates of SOP & POS.

1. XNOR Gate: It takes 2 or more input signals and output one signal.

The output signal is high or 1 or true, iff we had **even** number of high or 1 or true in a row of input signals, otherwise output is low or 0 or false.

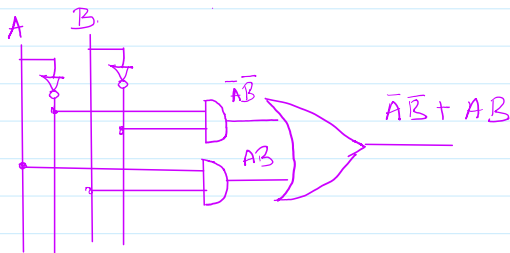
It is represented by '⊙' dot in a circle symbol.



A	B	A ⊙ B	
0	0	1	← $\bar{A} \cdot \bar{B}$ (Sum of Product) SOP
0	1	0	← $A + \bar{B}$ POS
1	0	0	← $\bar{A} + B$ POS (Product of sum)
1	1	1	← AB

SOP → $A \odot B = \bar{A}\bar{B} + AB$

POS → $A \odot B = (A+B)(\bar{A}+\bar{B})$



HW: Draw the logic gate ckt. for XNOR gate using the expression POS.

XNOR Gate using basic gates.

SOP Expression.

	A	B	C	A ⊙ B ⊙ C
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

1 is selected as output
0 is barred / complemented
SOP

$$A \odot B \odot C = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C + AB\bar{C}$$

POS ← 1 is barred / complement
0 is selected as output

$$A \odot B \odot C = (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+B+C)(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)(\bar{A}+\bar{B}+\bar{C})$$

5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

0 is selected as output

$$A \oplus B \oplus C = (A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)(\bar{A}+\bar{B}+\bar{C})$$

HW. Draw the logicckt. using basic gates of SOP & POS.

Boolean Algebraic Laws

1. Logical Addition(OR)

- i. $0 + 0 = 0$
- ii. $0 + 1 = 1$
- iii. $1 + 0 = 1$
- iv. $1 + 1 = 1$

2. Logical Multiplication

- i. $0.0 = 0$
- ii. $0.1 = 0$
- iii. $1.0 = 0$
- iv. $1.1 = 1$

3. Complement of bits

- i. $0' = 1$
- ii. $1' = 0$

4. Laws of boolean algebra

Law	Properties
$0 + p = p$ $0.p = p$	0
$1 + p = 1$ $1.p = p$	1
$p + p = p$ $p.p = p$	Idempotence law
$p + p.q = p$ $p + (p + q) = p + q$	Absorption law
$(p')' = p$	Involution law
$p + p' = 1$ $p.p' = 0$	Complementarity law
$(p + q) + r = p + (q + r)$ $(p.q).r = p.(q.r)$	Associative law
$p(q + r) = p.q + p.r$ $p + q.r = (p + q)(p + r)$ $p + p'.q = p + q$	Distributive law
$(p + q)' = p'.q'$ $(p.q)' = p' + q'$	Demorgan's law
$p \Rightarrow q = p' + q$	Implication or Conditional Elimination or if...then
$(p \Leftrightarrow q) = (p \Rightarrow q).(q \Rightarrow p) = p'.q' + pq$	Bi-conditional elimination or equivalence or

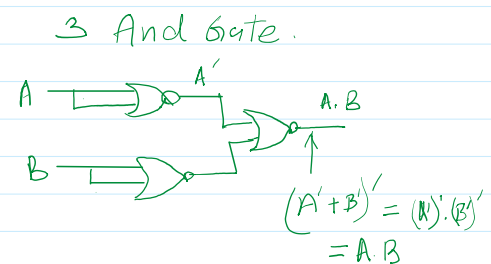
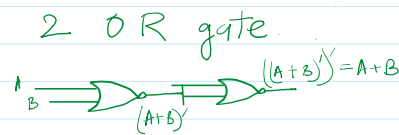
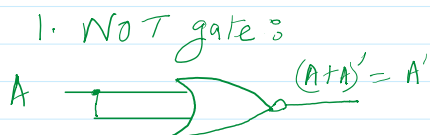
$(p \Leftrightarrow q) = (p \Rightarrow q) \cdot (q \Rightarrow p) = p' \cdot q' + pq$	Bi-conditional elimination or equivalence or if and only if (iff)
--	--

p	q	p'	q'	(p+q)'	p'.q'	(p.q)'	p' + q'
0	0	1	1	1	1	1	1
0	1	1	0	0	0	1	1
1	0	0	1	0	0	1	1
1	1	0	0	0	0	0	0

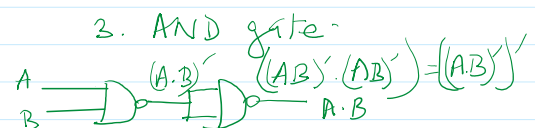
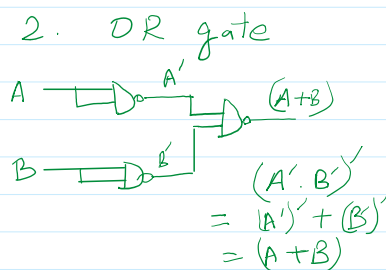
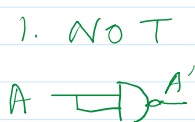
$(p+q)' = p' \cdot q'$ $(p \cdot q)' = p' + q'$

What is truth table?
 A truth table is a systematic way to represent all possible combination of inputs and their corresponding outputs for a logical expression or circuit. It is widely used in Boolean Algebra, digital logic and design, and computer science to analyze and validate logical operations.

NOR to NOR logic: Using NOR gates only we can design any type of basic gates.



NAND to NAND logic: Using NAND gates only we can draw all type of basic gates.



Why NAND and NOR Gates are called Universal Gates?

Ans: We can design ckts. using AND, OR and NOT gates but NAND and NOR gates are more popular as these are less expensive and easier to design. And also other switching functions (AND, OR, NOT, XOR, XNOR) can be easily implemented using NAND/NOR gates.

Thus NAND and NOR gates are also referred to as universal gates.

can be easily implemented using NAND/NOR gates.

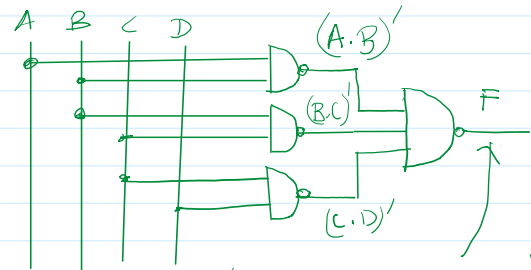
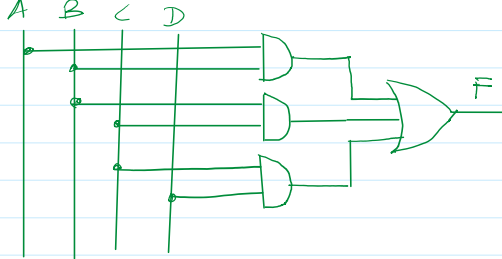
Thus NAND and NOR gates are also referred to as universal gates.

Skill:

Design rule for NAND to NAND logic network:

1. Draw simplified sum of product or SOP expression.
2. Draw the circuit diagram using AND, OR, NOT gates or basic gates.
3. Just replace AND OR, NOT gates with NAND gates.

$$F = AB + BC + CD$$

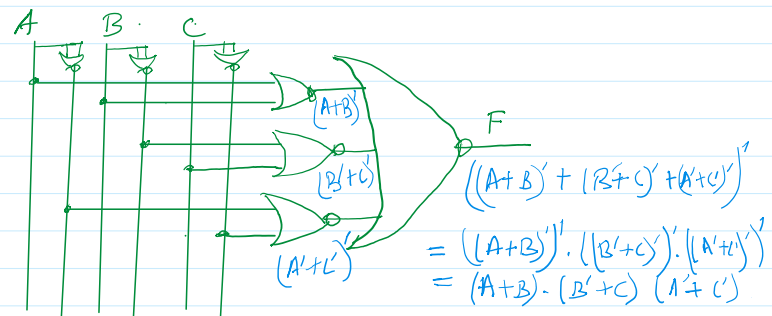
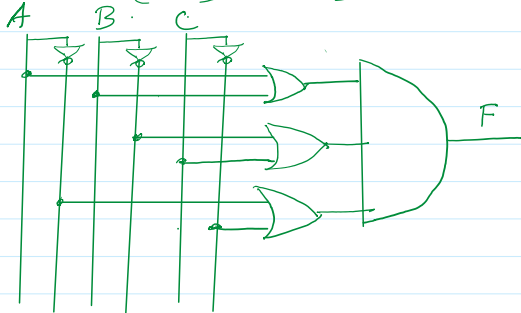


$$\begin{aligned} &= ((AB)' \cdot (BC)' \cdot (CD)')' \\ &= (AB)' + (BC)' + (CD)' \\ &= AB + BC + CD \end{aligned}$$

Design rule for NOR to NOR logic network:

1. Draw simplified sum of product or POS expression.
2. Draw the circuit diagram using AND, OR, NOT gates or basic gates.
3. Just replace AND OR, NOT gates with NOR gates.

$$F = (A+B) \cdot (B'+C) \cdot (A'+C')$$



$$F = (A+B) \cdot (B'+C) \cdot (A'+C') \quad \text{Draw the ckt. using NAND gates only.}$$

↓
dual

$$F_1 = A \cdot B + B' \cdot C + A' \cdot C' \Rightarrow \text{Apply NAND to NAND logic.}$$

Home work:

1. Draw XOR & XNOR (2 input & 3 input) circuits using

(i) NAND to NAND logic

(i) NAND to NAND logic

(ii) NOR to NOR logic